

Syllabus of Databricks Spark Certification Training

Module 1: Spark Architecture Components

Driver

Executor

Cores/Slots/Threads

Partitions

Module 2: Spark Execution

Jobs

Stages

Tasks

Module 3: Spark Concepts

Caching

Shuffling

Partitioning

Wide vs. Narrow Transformations

DataFrame Transformations vs. Actions vs. Operations

High-level Cluster Configuration

Module 4: SparkSession

Create a DataFrame/Dataset from a collection (e.g. list or set)

Create a DataFrame for a range of numbers

Access the DataFrameReaders

Register User Defined Functions (UDFs)

Module 5: DataFrameReader

Read data for the “core” data formats (CSV, JSON, JDBC, ORC, Parquet, text and tables)

How to configure options for specific formats

How to read data from non-core formats using format() and load()

How to specify a DDL-formatted schema

How to construct and specify a schema using the StructType classes

Module 6: DataFrameWriter

Write data to the “core” data formats (csv, json, jdbc, orc, parquet, text and tables)

Overwriting existing files

How to configure options for specific formats

How to write a data source to 1 single file or N separate files

How to write partitioned data

How to bucket data by a given set of columns

Module 7: DataFrame (Dataset)

Know how to cache data, specifically to disk, memory or both

Know how to uncache previously cached data

Converting a DataFrame to a global or temp view

Applying hints

Module 8: Spark SQL Functions

Elicitation and collaboration Knowledge Area

Exploring BRD Document

Stakeholder Analysis and Management

Requirement Elicitation Techniques

Module 9: Operations

Aggregate functions

Collection functions

Date time functions

Math functions

Misc functions

Non-aggregate functions

Sorting functions

String functions

UDF functions

Window functions